

1(a)(i)	glycerol ;	1
1(a)(ii)	any one from: hydroxyl / polar, group, interacts with, phosphate heads ; A idea that both are, polar / hydrophilic A hydroxyl group, faces aqueous environment / AW, as it is, polar / hydrophilic non-polar part, in region of / AW, fatty acid tails / AW, as both are, non-polar / hydrophobic ; A non-polar part is hydrophobic so, in centre of membrane / away from aqueous environment	1
1(a)(iii)	any one from: maintains / regulates, fluidity of membrane ; A detail e.g. reduces fluidity at high temperature / increases fluidity at low temperature / AW maintains / regulates, (mechanical) stability of membrane ; AW prevents entry of, hydrophilic substances / polar substances / ions ; without cholesterol membranes would easily rupture flat ring (<i>structure</i>) interferes with the movement of fatty acid tails reduces lateral movement of phospholipids	1
1(b)(i)	sodium ions are (positively charged), so repelled / AW, by the, hydrophobic tails / non-polar core / non-polar tails / AW ;	1

Question	Answer	Marks								
1(b)(ii)	<i>similarity:</i> any one from 1 both occur through / involve, a, membrane / transport, protein ; A carrier protein <i>if context is correct</i> 2 (both can be) specific to the, molecule / ion (passing through) ; A correct ref. to binding site(s) on protein 3 (both involve) conformational change of (carrier) protein) ; R if incorrect context of channel protein AVP ; e.g. can transport substances, into and out of cell / in both directions ; <i>differences:</i> <table><tr><td><i>facilitated diffusion</i></td><td><i>active transport</i></td></tr><tr><td>substances transported down the concentration gradient</td><td>substances transported against a concentration gradient</td></tr><tr><td>passive A does not require, ATP / (metabolic) energy</td><td>requires, ATP / (metabolic) energy</td></tr><tr><td>involves channel and carrier proteins</td><td>does not involve channel proteins / only involves carrier proteins</td></tr></table> <i>If not given as a similarity allow as differences</i> <i>idea that active transport always involves conformational change (facilitated diffusion only carrier protein)</i> <i>idea that active transport always involves specific binding site(s) (facilitated diffusion only with carrier proteins)</i>	<i>facilitated diffusion</i>	<i>active transport</i>	substances transported down the concentration gradient	substances transported against a concentration gradient	passive A does not require, ATP / (metabolic) energy	requires, ATP / (metabolic) energy	involves channel and carrier proteins	does not involve channel proteins / only involves carrier proteins	3
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1(c)(i)	any one from: SER (membrane-bound so) can provide, phospholipid / arachidonic acid / substrate ; (prostaglandins are lipids and) SER is involved in lipid transport ; prostaglandins can be transported (from SER) or stored (in SER) ; idea of compartmentalisation / separated from other reactions in the cytoplasm / provides optimum conditions for pathway / higher concentration of enzymes (of pathway) ;	1								

Question	Answer	Marks
1(c)(ii)	any three from: ref. to effect on, ionic bonding / hydrogen bonding / hydrophobic interactions, (with other R-groups) ; changes the, shape / conformation, of the active site ; active site no longer complementary to, arachidonic acid / substrate ; ref. to effect of activation energy not being reduced ; e.g. ref to changed charges (so no electron transfer) no longer provides hydrophobic regions for reaction to occur enzyme–substrate complex, not formed / formed at reduced rate ;	3
1(c)(iii)	any two from: A ligands for prostaglandins <i>if in correct context</i> 1 prostaglandins are, secreted / released, by cells or prostaglandins are, transported / AW, to target cells ; 2 prostaglandins bind to receptors (on target cell surface membranes) ; R antigens 3 example of events triggered leading to a response ; e.g. activation of secondary messenger enzyme, cascade / activation phosphorylation events signal transduction	2